

Electronic Thermometer

Circuit Design and PCB Layout Report

Layout of ICs and PCBs — Exam Topic 3

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Fully analog design · single +5 V rail · no programmable logic

1 Introduction and Objective

This report documents the design and printed-circuit-board (PCB) implementation of an electronic thermometer built entirely from discrete transistors and standard analog and logic integrated circuits, with no microcontroller or programmable logic. The measured temperature is displayed on six range LEDs, and an audible alarm sounds when the highest temperature range is reached.

The complete signal chain is: an LM235 temperature sensor, a difference-amplifier conditioning stage, a six-stage discrete-transistor decoder driving the LEDs, and a gated two-timer oscillator that produces the alarm tone. The entire circuit runs from a single +5 V supply, and every internal node is made accessible to an Advantest 93000 chip tester through two 2×30 pin-socket connectors. The design was captured and laid out in KiCad 10 and verified in the ngspice simulator.

Key requirements met: single +5 V supply; no microcontrollers or programmable logic; a discrete-transistor display decoder; full tester access with all even connector pins grounded and forced cross-currents held below 1 mA; a ground copper pour on both layers; and a two-layer board within the permitted size.

2 System Overview

Figure 1 shows the block diagram. The LM235 produces a voltage proportional to absolute temperature. The difference amplifier subtracts a fixed reference and applies gain, spreading the narrow sensor window across a wider range so the decoder can resolve it cleanly. The decoder lights the six LEDs in sequence as the temperature rises. When the Red 2 (highest) stage is reached, its signal is combined in an AND gate with a gated 1 kHz oscillator output, and the result drives the buzzer.

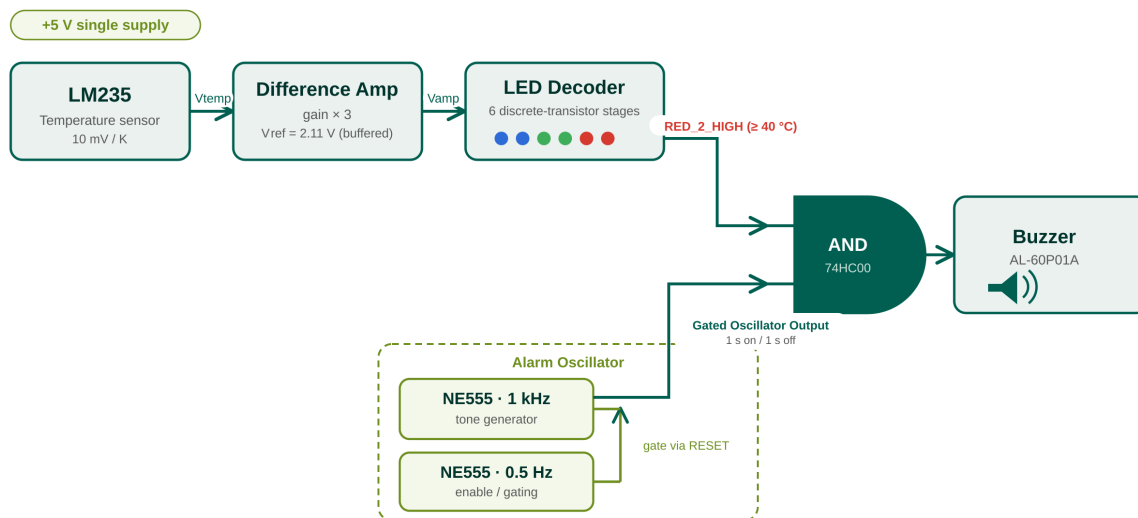


Figure 1 — System block diagram: sensor → amplifier → decoder, with the Red 2 threshold and the gated oscillator combined in the alarm logic to drive the buzzer.

3 Temperature Sensor (LM235)

The LM235 is a precision temperature sensor whose terminal voltage rises 10 mV per kelvin. It is used as a two-terminal device, biased from the +5 V rail through a 2.2 k Ω series resistor (R1). This sets an operating current of approximately $(5\text{ V} - V_{\text{sensor}})/R1 \approx 0.9\text{ mA}$ at room temperature, comfortably inside the LM235's 0.4–5 mA operating window across the whole $-10\text{ }^{\circ}\text{C}$ to $+40\text{ }^{\circ}\text{C}$ range. At $25\text{ }^{\circ}\text{C}$ (298 K) the sensor sits at about 2.98 V.

Because the sensor has a relatively high source impedance, drawing current from it directly would cause its output to sag and corrupt the reading. It is therefore never loaded directly by the decoder; its output is buffered by the amplifier stage (high input impedance), which presents negligible load to the sensor. The sensor node is also routed to the tester through a CD4066 analog switch so it can be isolated and driven during test.

4 Amplifier Stage

The raw sensor window spans only about 500 mV over the full temperature range ($\approx 2.63\text{--}3.13\text{ V}$), sitting on a large DC pedestal. If fed directly to the decoder, the six switching thresholds would be crammed into that narrow band and would switch softly. A difference amplifier (one half of an LM358, U2A) solves this by subtracting a fixed reference and applying a gain of three:

$$V_{\text{amp}} = (R_f / R_{\text{in}}) \cdot (V_{\text{temp}} - V_{\text{ref}}) = 3 \cdot (V_{\text{temp}} - 2.11\text{ V})$$

with $R_f = 30\text{ k}\Omega$ and $R_{\text{in}} = 10\text{ k}\Omega$. The reference $V_{\text{ref}} = 5\text{ V} \cdot R_{16}/(R_{15}+R_{16}) = 5 \cdot 10/(13.7+10) = 2.11\text{ V}$. Because a plain resistor divider has a significant source (Thévenin) resistance of about 5.8 k Ω , it would unbalance the difference amplifier's input arms; the reference is therefore buffered by the second half of the LM358 (U2B, unity gain) so it presents a stiff, low-impedance 2.11 V.

Placing V_{temp} on the non-inverting input means the sensor sees only about 75 μA of loading — negligible — so this single stage both buffers the sensor and stretches the signal. The output remains within the single-supply range (peaking near 3.4 V, below the LM358 ceiling). The result is that the six thresholds fan out across roughly 1.5 V instead of 0.5 V, giving crisp, well-separated switching.

5 LED Decoder

5.1 Concept

The decoder is six identical common-emitter NPN stages (Q9–Q14), one per LED — a discrete-transistor resistor-transistor-logic (RTL) design, as the specification requires. Each transistor's base is fed by a resistor divider from V_{amp} ; a slightly different divider ratio makes every stage switch on at a different V_{amp} , and therefore at a different temperature. A 1N4148 diode in each emitter (D2–D7) raises the turn-on voltage and, more importantly, gives each stage an independent emitter reference so that several lit LEDs do not clamp one another's turn-on point. As the temperature rises the LEDs light in sequence, and colder LEDs remain lit — which the task permits.

5.2 Threshold Equation and Values

A stage conducts when its base reaches the turn-on voltage $V_{\text{on}} = V_{\text{be}} + V_{\text{diode}} \approx 1.4\text{ V}$. With the base voltage given by the divider, $V_{\text{base}} = V_{\text{amp}} \cdot R_{\text{bot}}/(R_{\text{top}}+R_{\text{bot}})$, the required top resistor for a target switch-on voltage is:

$$R_{\text{top}} = R_{\text{bot}} \cdot (V_{\text{amp_th}} / V_{\text{on}} - 1)$$

Worked example (Red 2, the alarm, at 40 °C): $V_{amp_th} = 3 \cdot (3.132 - 2.11) = 3.07 \text{ V}$, so $R_{top} = 10 \text{ k}\Omega \cdot (3.07/1.4 - 1) \approx 11.8 \text{ k}\Omega$. R_{bot} is kept at 10 k Ω on every stage: this keeps the total current drawn from Vamp light (about 2 mA across all six dividers), so the amplifier holds Vamp stiff and the stages do not interact through supply droop.

The six stages, their target temperatures, the resulting Vamp thresholds and the divider values are summarised in Table 1.

Table 1 — Decoder switching thresholds (gain 3, $V_{ref} = 2.11 \text{ V}$, $V_{on} \approx 1.4 \text{ V}$, $R_{bot} = 10 \text{ k}\Omega$).

LED	Temp	Vtemp	Vamp (switch)	Rtop	Rbot
Blue 1	−10 °C	2.632 V	1.57 V	1.18 k Ω	10 k Ω
Blue 2	5 °C	2.782 V	2.02 V	4.42 k Ω	10 k Ω
Green 1	15 °C	2.882 V	2.32 V	6.49 k Ω	10 k Ω
Green 2	25 °C	2.982 V	2.62 V	8.66 k Ω	10 k Ω
Red 1	30 °C	3.032 V	2.77 V	9.76 k Ω	10 k Ω
Red 2	40 °C	3.132 V	3.07 V	11.8 k Ω	10 k Ω

5.3 Switching Behaviour

Because the amplifier spreads the window, adjacent thresholds are 150–450 mV apart — wider than a bipolar transistor's soft turn-on band of roughly 120 mV — so each LED switches cleanly before the next begins, avoiding overlap. On the collector side the LED node is a two-level node: about 5 V when the stage is off (pulled up through the LED and its anode resistor) and about 1.4 V when on. The on-level is not 0 V because it is set by $V_{ce(sat)}$ plus the emitter-diode drop; this floor is expected and does not affect the display.

6 Alarm Subsystem

6.1 Gated Two-Timer Oscillator

The alarm tone is generated by two NE555 timers in astable mode. A fast timer produces the $\approx 1 \text{ kHz}$ audible tone; a slow timer produces a $\approx 0.5 \text{ Hz}$ enable signal. The slow timer gates the fast one through its RESET pin, so the tone is switched on and off in roughly one-second bursts. The astable frequency for each stage is:

$$f = 1.44 / ((Ra + 2 \cdot Rb) \cdot C)$$

For the tone stage, $R_a = 1 \text{ k}\Omega$, $R_b = 6.8 \text{ k}\Omega$ and $C = 100 \text{ nF}$ give $f \approx 986 \text{ Hz}$. For the enable stage, $R_a = 1 \text{ k}\Omega$, $R_b = 143 \text{ k}\Omega$ and $C = 10 \text{ }\mu\text{F}$ give $f \approx 0.50 \text{ Hz}$, i.e. approximately one second on and one second off. A crystal oscillator was rejected as unnecessary — it would run at MHz and need a divider chain to reach an audio tone, adding parts and precision that a buzzer does not require — and the 555 meets the single-supply, fixed-function constraints directly.

6.2 Custom NE555 SPICE Model

The vendor NE555 models available were unreliable in ngspice, so a behavioural subcircuit was built from the timer's functional blocks: the internal 5 k–5 k divider (setting the $\frac{1}{3}$ and $\frac{2}{3}$ V_{cc} comparator references), two comparators, a NOR-based set-reset latch, and a discharge switch. A small RC on each latch node breaks the algebraic feedback loop and gives clean convergence. The model was validated

against the design targets, producing 972 Hz for the tone stage and 0.495 Hz (≈ 1.16 s on / 0.86 s off) for the enable stage.

6.3 Alarm Logic and Buzzer

Two conditions must both be true for the alarm to sound: the temperature must be at the Red 2 threshold (the highest range), and the gated oscillator must be active. A 74HC00 gate combines these, and its output drives the buzzer through a transistor stage. The AL-60P01A buzzer is driven by a low-side NPN switch (Q4) with a 1N4148 flyback diode (D10) across the inductive transducer and a series resistor limiting the drive current. The transistor isolates the buzzer's larger switching current from the logic — the buzzer current flows from the +5 V rail through the transistor to ground, not through the gate — and the flyback diode clamps the coil's turn-off spike. Below the Red 2 threshold the gate output is idle and the buzzer is silent.

7 Testability

Two 2×30, 2.54 mm pin-socket connectors (JM1 and JM2) give the Advantest 93000 access to the internal signals. All even-numbered pins are tied to ground, and about 21 signals are brought out (13 on JM1, 8 on JM2), within the 30-per-connector budget. Two isolation strategies are used so the tester can force nodes without excessive cross-current:

- Digital drive lines use 5.1 k Ω series resistors, so a forced node produces at most $5\text{ V} / 5.1\text{ k}\Omega \approx 1\text{ mA}$ of cross-current — within the specified limit.
- The analog nodes Vtemp and Vamp are routed through a CD4066 transmission gate (U4). With the gate closed the tester senses the node through the switch's low on-resistance; with the gate open, the sensor or amplifier is fully disconnected, so the tester can drive the node with no contention and therefore no cross-current at all. The SELECT control lines set each gate and are themselves current-limited.

8 PCB Layout

The board is a two-layer design measuring 104.14×76.2 mm (the larger permitted size, set by the two full-height tester connectors). The connectors JM1 and JM2 are placed on the back at the fixed template positions; all other components are on the front, with the power screw terminal in the marked top area and parts grouped by subsystem to keep nets short.

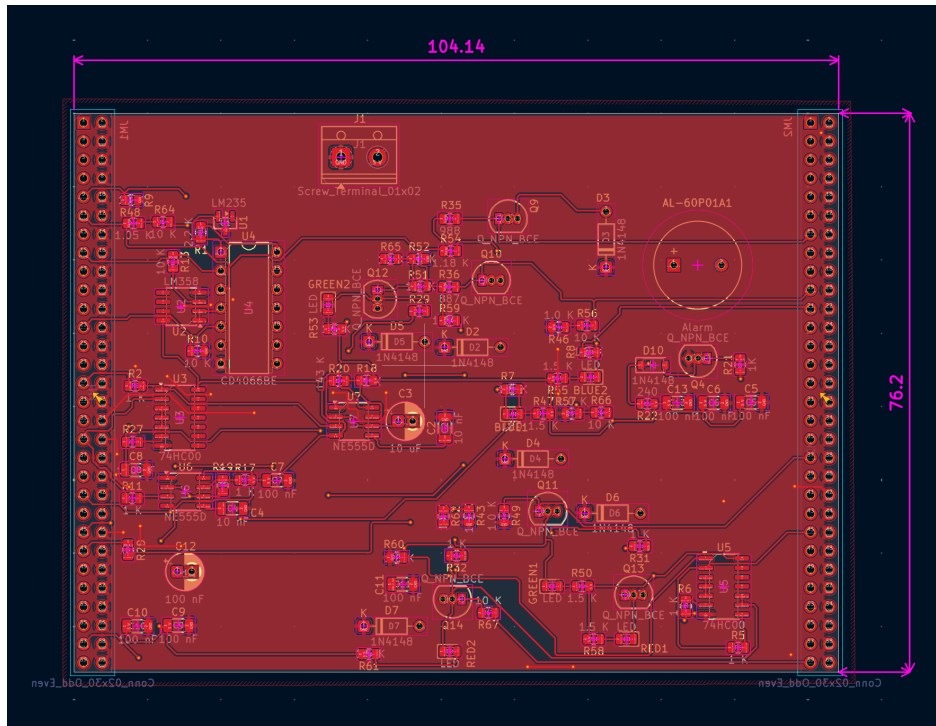


Figure 2 — PCB layout: front-copper routing and ground pour (board 104.14 × 76.2 mm).

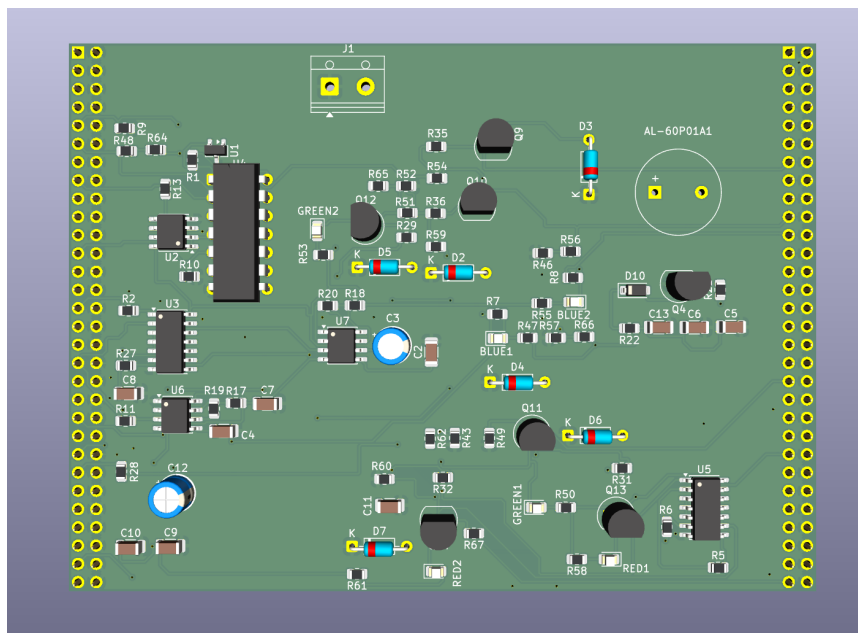


Figure 3 — 3D render of the assembled board (front side).

8.1 Footprints

All 88 board components use packages at or above the 0805 / SC70 minimum, and no package hides its pads, so every joint remains inspectable. The through-hole parts are the mechanically-loaded and inherently through-hole items. Table 2 lists the packages used.

Table 2 — Footprints used on the board.

Package	Type	Qty	Function
R 0805	SMD	45	dividers, series and drive resistors
C 1206	SMD	10	decoupling / timing
LED 0805	SMD	6	2 blue · 2 green · 2 red
SOIC-14 (74HC00)	SMD	2	alarm logic gating
SOIC-8 (LM358, NE555)	SMD	3	buffer and two timers
TO-92	THT	8	LM235 and Q4, Q9–Q14
DIP-14 (CD4066)	THT	1	analog transmission gate
Diode DO-35 (1N4148)	THT	6	emitter diodes D2–D7
Radial electrolytic	THT	2	NE555 timing capacitors
Pin socket 2×30	THT	2	JM1 / JM2 tester (back side)
Screw terminal 1×2	THT	1	+5 V input
Buzzer AL-60P01A	THT	1	audible alarm

8.2 Grounding and Copper Zones

A single ground zone covers the whole board on both the front and back layers, as required. The zone clearance was set to 0.2 mm and the minimum copper width to 0.15 mm so the pour floods cleanly into the tight 2.54 mm-pitch spaces between the tester-connector pads, giving a solid ground reference right up to the connectors. The front and back planes are tied together by ground stitching vias distributed across the interior, and the even-numbered connector pins act as further grounded through-holes that stitch the planes along both edges — keeping the ground a single low-impedance reference with no isolated copper. A continuous ground plane is important because the fast-switching logic ICs need a short, low-inductance return path for their decoupling capacitors to be effective and for the decoder thresholds to sit on a stable reference.

8.3 Trace Width and Routing

- Uniform track width of 0.2 mm, comfortably above the 0.125 mm minimum.
- 45° corners rather than single 90° bends, with trace-to-pad and trace-to-trace inner angles kept at 90° or more to avoid acid traps during etching.
- Teardrops at pad junctions to strengthen connections and smooth the entry angle.
- Thermal-relief spokes where tracks and the pour meet pads, to ease soldering; remaining empty areas filled with grounded copper.

Routing followed the priority order of short obvious nets first, then power and decoupling connections with short low-impedance paths, then the remaining signals. Both copper layers carry signals, with vias where a net must change side. There are 94 vias in total (0.6 mm pad, 0.3 mm drill), combining signal layer-changes with ground stitching.

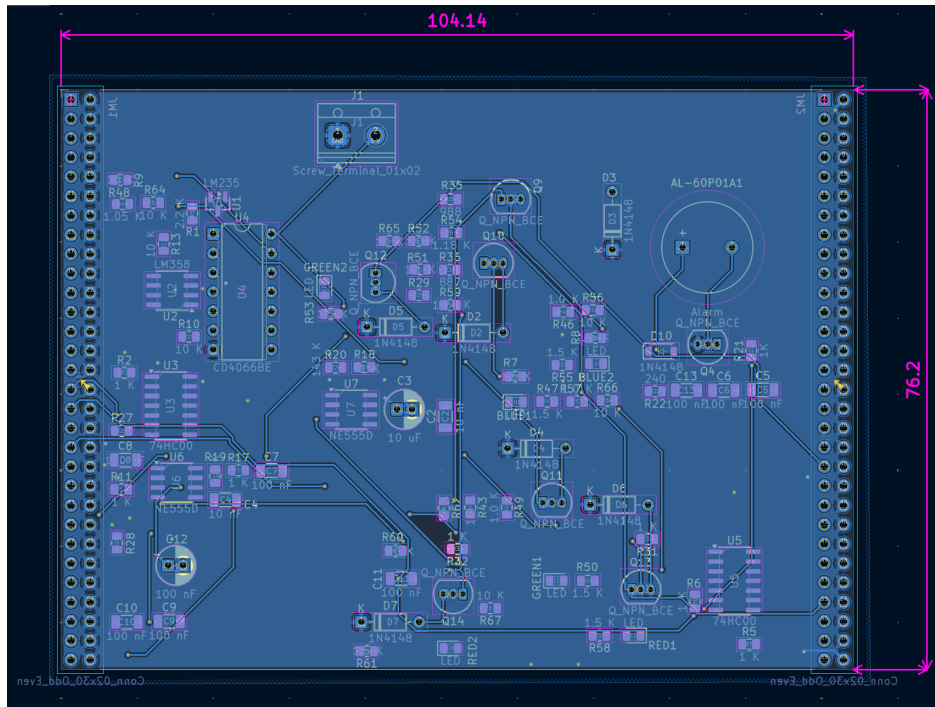


Figure 4 — Component placement (subsystem grouping; connectors on the back edge).

8.4 Design-Rule Checks and Footprint Validation

The layout passes the KiCad design-rule check with no errors, and all footprints were verified against their datasheets. A parity check against the hierarchical schematic confirms that every board component matches the schematic by reference, value and footprint; the simulation-only elements are correctly excluded from the layout. Table 3 summarises the key copper and via rules against the values used.

Table 3 — Design rules versus specification.

Rule	Specification (min)	Value in design
Track / clearance	0.125 mm	0.2 mm
Copper-to-hole clearance	0.25 mm	≥ 0.25 mm
Copper-to-edge clearance	0.30 mm	0.30 mm
Via drill diameter	0.25 – 0.45 mm	0.30 mm
Via annular ring	0.10 mm	0.15 mm
PTH annular ring	0.30 mm	0.35 mm (connectors)
GND zone on both layers	required	present, full board

9 Simulation Results

The circuit was verified in ngspice with both a DC sweep and a transient run. The DC sweep confirmed that the six LEDs switch on in order across the Vamp range. The transient run used a piecewise-linear source to imitate a temperature that rises and then falls: as Vamp climbed the LEDs lit in sequence, at the peak the Red 2 stage triggered and the gated oscillator produced 1 kHz bursts switched on and off at

about 0.5 Hz, and as the temperature fell the LEDs switched off again in reverse order. This confirms the full chain from sensor to display to alarm.

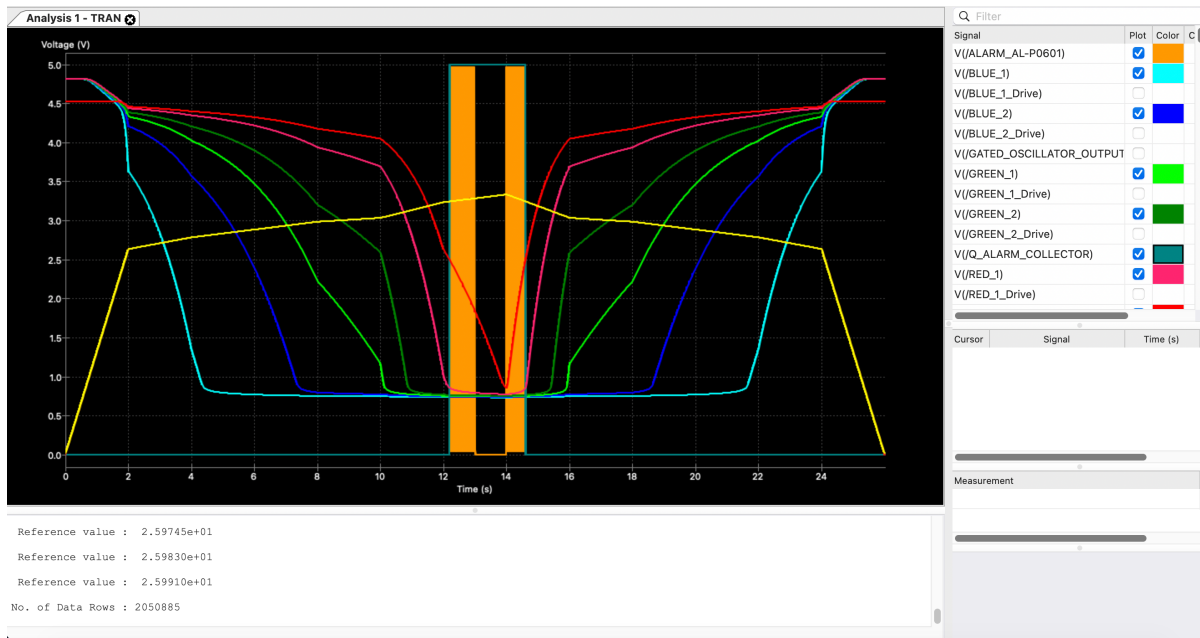


Figure 5 — Transient simulation: LED decoder outputs rising then falling, with the gated 1 kHz alarm burst at the highest temperature.

A practical note on simulation models: the switch-on point depends on the transistor and emitter-diode turn-on voltage. Generic default device models overstate this and shift the upper thresholds; assigning realistic device models (a standard NPN and 1N4148) brings all six stages into order within about $\pm 1^\circ\text{C}$ of their targets. As the top stage sits at the very top of the Vamp range, the alarm is best exercised by ramping a few degrees past 40°C so Red 2 crosses its threshold with margin.

10 Design Evaluation and Possible Improvement

The RTL decoder meets the specification and works, but its switching threshold is set by the transistor's own base-emitter (plus emitter-diode) voltage. That voltage is an absolute, temperature-dependent and part-dependent quantity, so the threshold drifts (about $-2\text{ mV}/^\circ\text{C}$) and the single transistor turns on softly over roughly 120 mV. The principled improvement, while remaining fully within the discrete-transistor rule, is to replace each stage with a two-stage comparator (Figure 6).

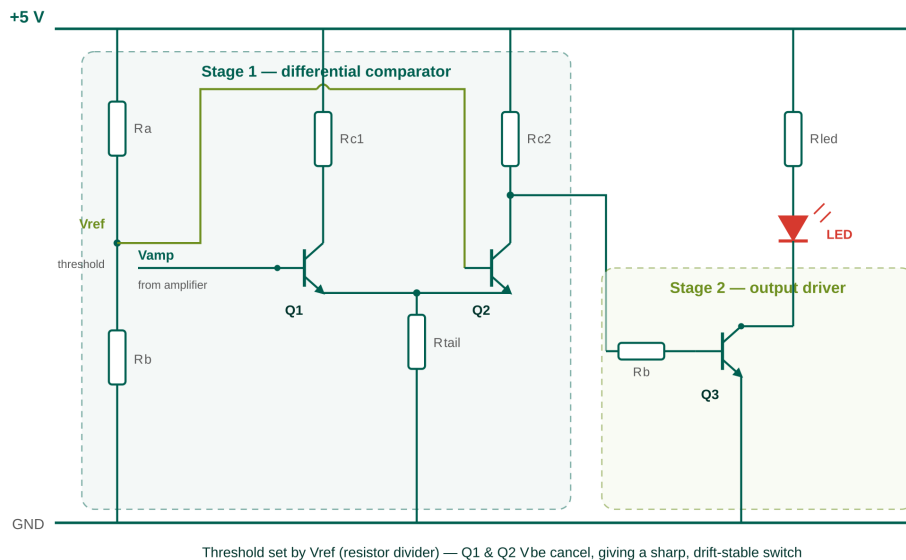


Figure 6 — Improved decoder stage: a differential-pair comparator (Stage 1) compares V_{amp} to a resistor-divided V_{ref} , and an output transistor (Stage 2) drives the LED.

Stage 1 is a differential pair sharing a fixed tail current, with V_{amp} on one base and a resistor-divided V_{ref} on the other. The ratio of the two transistor currents depends only on $(V_{amp} - V_{ref})/V_T$, so the pair balances — and the output flips — exactly at $V_{amp} = V_{ref}$. This gives three advantages:

- The threshold is set by V_{ref} (resistors, accurate to a few mV), not by the transistor's V_{be} .
- The two matched transistors' V_{be} values cancel, so the temperature drift is removed to first order.
- The differential pair has gain ($g_m \cdot R_c$), so it switches sharply over tens of millivolts instead of ~ 120 mV — eliminating overlap between adjacent LEDs.

Stage 2 is a common-emitter output transistor that drives the LED cleanly and keeps the LED current off the reference, so stages no longer interact. A single feedback resistor adds hysteresis (Schmitt action) to eliminate flicker at the switch point. The trade-off is roughly two to three transistors per stage plus a standing tail current, so more board area — best implemented with matched dual-transistor packages. This is presented as the recommended refinement rather than a change to the delivered board.

11 Conclusion

The electronic thermometer meets the task requirements as both a circuit and a physical board. It senses temperature with the LM235, conditions the signal with a buffered difference amplifier, indicates it on six range LEDs using discrete-transistor stages, and produces a gated 1 kHz alarm at the highest threshold — all from a single +5 V rail and without any programmable logic. Full chip-tester access is provided through two 2×30 connectors with all even pins grounded, digital lines isolated by 5.1 kΩ series resistors and the analog nodes isolated by a transmission gate. The board is a DRC-clean, two-layer design with a full ground pour on both sides, and the complete sensor-to-alarm chain has been verified in simulation.